

SUMMARY TABLE: CORE MATHEMATICS GRADE 10

Sub-Strand 2.3: Rotation — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.3: Rotation
Driving Question	DRIVING QUESTION: When the minute hand of the clock at the University of Nairobi completes its journey around the clock face, what mathematical rules govern exactly where every point on the hand ends up — and how can we use those rules to rotate any shape to any position, on paper and on a grid?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 The Spinning Hands of Nairobi's Clock Tower: What Is Really Happening?	Learners observed the minute hand of the University of Nairobi Administration Block clock at four positions — 12:00, 12:15, 12:30, and 12:45 — and noted that the hand's length and shape remained identical at every position while its direction changed completely.	Learners discovered, through the improvised card-arrow clock investigation, that every point on a rotating object stays the same distance from the centre of rotation — and that this type of motion (rotation) is distinct from a translation (slide) or a reflection (flip). Key vocabulary introduced: centre of rotation, angle of rotation, clockwise, anticlockwise.	Learners explained that the clock hand is performing a rotation about a fixed centre point, that no stretching or flipping occurs, and that the distance from the centre to any point on the hand is preserved. They recorded this as a boxed property in their Rotation Model Journals and posted their initial questions and first partial answers on the class Driving Question Board.
Lesson 2 Discovering the Properties of Rotation	Learners physically rotated paper cutouts of triangles and quadrilaterals about a fixed pin through angles of 90° , 180° , and 270° in both clockwise and anticlockwise directions, tracing object and image positions on A3 plain paper and recording measurements of corresponding sides, angles, distances from centre (OA, OA', OB, OB'), and angular displacements (angle AOA', BOB', COC') in a structured group observation table.	Learners discovered and co-constructed the four properties of rotation: (1) the object and image are congruent — all corresponding sides and angles are equal; (2) the centre of rotation is invariant — it is the only point that does not move; (3) every point on the object moves through exactly the same angle about the centre; (4) the direction of rotation (clockwise vs. anticlockwise) determines where the image ends up, so direction must always be specified. These properties were connected to the movement of the University of Nairobi clock minute hand (the anchoring phenomenon) and to the rotating millstone of a posho mill in	Learners updated their cumulative annotated sketch model of the University of Nairobi clock, adding colour-coded arcs showing equal 90° angular displacement from multiple points on the minute hand, labelling the clock centre as 'invariant', recording the four properties in a box, and adding a posho mill millstone sketch with its axle labelled as 'centre of rotation'. Learners also updated the Driving Question Board with evidence notes (green) linking today's findings to prior DQB questions and new wonder notes (orange) pointing toward how exact image coordinates can be calculated — the focus of Lesson 3.

		Kericho (the supporting phenomenon).	
Lesson 3 Rotating Objects on a Plane Surface — Constructing Images with a Geometrical Set	Learners observed that using a compass pin at the fixed centre O, a compass arc preserving the distance from O to each vertex, and a protractor measuring the given angle produces a precise image of any polygon — and that changing the position of the centre (inside, outside, or on the boundary of the object) changes where the image lands but does not change the four-step construction method.	The four-step construction method for rotating an object on a plain surface: (1) mark the centre of rotation; (2) draw rays from the centre through each vertex; (3) use a compass to preserve the distance from centre to each vertex; (4) use a protractor to measure the specified angle at the centre and mark the image vertex. The centre is always the only invariant point, regardless of its position relative to the object.	The construction method directly models what the University of Nairobi clock's minute hand does: the compass pin represents the fixed clock centre, the compass arc represents the constant length of the hand from axle to tip, and the protractor angle represents the angular sweep of the hand. Constructing the hand at 12:15 (90° clockwise) and 12:45 (270° clockwise) on the clock model showed that the same four-step method predicts the exact position of every point on the hand — beginning to answer the driving question about the mathematical rules governing where every point ends up.
Lesson 4 Rotation on the Cartesian Plane — Discovering and Applying Coordinate Rules	Learners plotted triangle ABC on a Cartesian grid and constructed its rotated images through 90° anticlockwise, 90° clockwise, and 180° about the origin using their geometrical sets. They recorded the before-and-after coordinates of all three vertices in a structured table and circled numerical patterns between the original and image coordinates. They also observed that the distance of each vertex from the origin is identical in the object and all three images, confirming the distance-preservation property of rotation on a coordinate grid.	From the coordinate tables, learners inductively discovered and then formalised four algebraic rules: 90° anticlockwise about the origin maps $(x, y) \rightarrow (-y, x)$; 90° clockwise maps $(x, y) \rightarrow (y, -x)$; 180° maps $(x, y) \rightarrow (-x, -y)$; and 270° anticlockwise gives the same result as 90° clockwise. They also learned the translate-apply-translate-back strategy for rotating about a centre other than the origin, and applied all rules to the Kasarani football context to calculate the new position of a marked point after successive rotations.	The coordinate rules explain precisely what happens to every point on the University of Nairobi clock minute hand as it rotates: each point's distance from the centre (the clock's pivot) is preserved — confirmed algebraically because $\sqrt{((-y)^2 + x^2)} = \sqrt{(x^2 + y^2)}$ — and its new position after any multiple of 90° can be calculated exactly without drawing. The minute hand tip at (0, 5) on a grid maps to (-5, 0) after 90° anticlockwise (12:45 position), (0, -5) after 180° (12:30), and (5, 0) after 270° anticlockwise (12:15), precisely matching the four clock positions from the anchoring phenomenon. Mathematical rules, not just construction, now govern where every point ends up.
Lesson 5 INVESTIGATE: Finding the Centre and Angle of Rotation	Learners observed that a triangle and its rotated image — with no centre or angle given — contain enough geometric information to reconstruct the rotation completely. They saw, through their own compass-and-ruler construction, that the perpendicular bisectors of two object-image point-pair segments always meet at a single point: the centre of rotation. They also observed that the angle measured at that centre between any object point and its image point is the same for all corresponding pairs.	Learners learned that: (i) the centre of rotation is equidistant from each object point and its corresponding image point, so it lies on the perpendicular bisector of every object-image segment; (ii) two perpendicular bisectors are sufficient to uniquely locate the centre; (iii) the angle of rotation is measured at the centre between the object ray and the image ray, using a protractor; (iv) the result can be verified by performing the rotation and checking it maps the object onto the image; (v) this reverse-engineering method works on both	The mathematical explanation is: for any rotation, every point P and its image P' are equidistant from the centre O (since rotation preserves distances). Therefore O lies on the perpendicular bisector of PP'. This is true for every corresponding pair, so the perpendicular bisectors of PP' and QQ' both pass through O — their intersection is therefore uniquely O. The angle of rotation is $\angle POP'$, which equals $\angle QOQ'$ for all corresponding pairs. This is a consequence of the isometric (distance-preserving) nature of rotation.

		plain paper and the Cartesian plane.	
Lesson 6 Rotational Symmetry of Plane Figures and Solids	Learners physically rotated paper cutouts of an equilateral triangle, a square, and a regular hexagon about a pin-held centre point and counted the number of times each shape mapped exactly onto its own outline in one full 360° rotation. They also examined and rotated models of a square pyramid, triangular prism, cone, and tetrahedron to identify each solid's axis and order of rotational symmetry. Results were compiled into a class table on the Driving Question Board.	A plane figure has rotational symmetry of order n if it maps onto itself exactly n times in a full 360° rotation about its centre. The smallest angle of rotation that achieves this is $360^\circ/n$. The equilateral triangle has order 3 (angle 120°), the square has order 4 (angle 90°), and the regular hexagon has order 6 (angle 60°) — matching the hexagonal weave pattern of a Maasai kiondo basket. Three-dimensional solids have an axis of rotational symmetry; the triangular prism has order 3 about its longitudinal axis, the square pyramid has order 4 about its vertical axis, the cone has infinite order about its vertical axis, and the tetrahedron has order 3 about an axis through one vertex and the centre of the opposite face. Every shape or solid has at least order 1.	The order of rotational symmetry reveals how many times a shape reproduces its own appearance during one complete rotation about a fixed centre or axis — exactly as the minute hand of the Nairobi University clock returns to an equivalent position. A regular polygon with n sides has rotational symmetry of order n because rotating it by $360^\circ/n$ maps every vertex to the next identical vertex position. This is a direct consequence of the equal side lengths and equal interior angles that define a regular polygon. The same principle applies to the cross-sectional face of a prism: a triangular prism's order-3 symmetry about its axis is determined by the equilateral triangle at each end. The kiondo basket's hexagonal pattern (order 6) and the Rift Valley windmill's three-blade arrangement (order 3) are real-world applications of this mathematical rule, and the full 12-mark Nairobi clock face is a real-world example of order-12 rotational symmetry with a 30° rotation angle between each coincident position.
Lesson 7 Rotation and Congruence: Same Shape, Different Story	Learners physically tested four pairs of object-and-image cutouts (translation, rotation, reflection, and the Nairobi clock hand) using a three-step superimposition protocol — slide, flip-then-slide, lift — and recorded which pairs required lifting to match.	Rotation produces OPPOSITE congruence: the rotated image is the same shape and size as the object, but its cyclic vertex order is reversed, so it cannot be superimposed on the object without lifting it off the plane. Translation produces DIRECT congruence (slide only). Reflection also produces opposite congruence. The minute hand of Nairobi's clock at different positions exemplifies opposite congruence through rotation.	The reversal of cyclic vertex order under rotation — clockwise becomes anticlockwise — is the geometric reason that rotation produces opposite congruence, even though no mirror is involved. This unifies the congruence landscape: direct congruence preserves orientation; opposite congruence reverses it, regardless of whether the reversal is caused by a rotation or a reflection.
Lesson 8 Model Building, Application & Assessment: "Putting It All Together"	Learners re-watched the anchoring phenomenon — the University of Nairobi Administration Block clock — and observed the minute hand (modelled as a line segment with points A(0,5), B(0,3), C(0,1) on a coordinate grid centred at the origin) at 12:00, 12:15, 12:30, 12:45, and 1:00. They observed that the length of the hand, the	Learners learned and applied all seven SLOs of Sub-Strand 2.3: (a) rotation on the Cartesian plane using rules — 90° CW: $(x,y) \rightarrow (y,-x)$; 180°: $(x,y) \rightarrow (-x,-y)$; 270° CW / 90° ACW: $(x,y) \rightarrow (-y,x)$; 360°: $(x,y) \rightarrow (x,y)$; (b) construction of the centre of rotation using perpendicular bisectors of segments joining object points to their images; (c)	The minute hand of the University of Nairobi clock is a real-world example of rotation: a geometric transformation in which every point of an object turns through a fixed angle about a fixed centre (the origin/clock centre), preserving all distances and angles. The mathematical rules that govern exactly where every point ends up are

	shape, and the distance of every point from the centre remained invariant at every position, while the direction of the hand changed by exactly 90° clockwise every 15 minutes.	order of rotational symmetry of plane figures (regular hexagon = 6, rectangle = 2, letter S = 2); (d) axis and order of rotational symmetry of a triangular prism solid; (e) deduction of congruence (direct congruence) from rotation; (f) real-life identification of rotation in Nairobi and Kenyan local contexts; (g) synthesis of all concepts into a full written answer to the driving question.	coordinate rotation rules: for a 90° clockwise rotation, $(x,y) \rightarrow (y,-x)$; for 180°, $(x,y) \rightarrow (-x,-y)$; for 270° clockwise, $(x,y) \rightarrow (-y,x)$. These rules allow us to predict the precise location of every point on any shape after any rotation on the Cartesian plane. The object and its image are always directly congruent (same shape, same size, same orientation type). The clock hand itself has rotational symmetry of order 1 (as a non-symmetric shape), while the clock face with its 12 hour-marks has rotational symmetry of order 12 (only a plain, unmarked circle would have infinite order) — both are explained by the same underlying mathematical structure of rotation.
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